

project published in September 2011 issue, I have following doubts:

1. In Fig. 2, the buzzer (PZ1) seems to have been connected in reverse polarity.

2. The starter activation circuit was recommend only for DPDT relay (RL2), but the starter circuit I am using (starter from CRI pumps) has six contacts.

3. In the auto cut-off section, SCR2 (BT136) is used, which is not functioning properly in my circuit. BT169 seems to be working fine as its gate is triggered, but T2 is not going to cut-off state.

4. In Fig. 5, the internal connection details are not given.

5. What type of sensors can be used for the level and flow monitor section?

Sudhakar V.

Through e-mail

The author B. Shankar Rao replies :

1. The polarity of buzzer was printed wrongly in Fig. 2. It should be reversed for proper operation of the buzzer.

2. Most submersible starter units have only four contacts (refer Fig. 5 in the article)—two for contactor and two for capacitor. I have not come across any six-contact starter or start button so far. If I get more information about the starter unit and full details about the pump (submersible or sump), I may be able to help.

3. SCR2 is used for automatic (timer) cutting-off/switching-off of the pump through relay RL1. This action takes place only when the sensor's wires are broken and the pump is not stopped when the overhead tank is full. This time selection (30 minutes, 60 minutes or 120 minutes as required) is done based on the capacity of the tank. The timer starts the moment pump-control relay (RL1) is activated. If the overhead tank gets filled before the selected time and RL1 de-energises, SCR2 doesn't activate.

In case the pump continues to run, SCR2 activates to switch off T2 and the pump stops. The switching on of SCR2 activates the buzzer circuit to warn that sensor wires are

broken or something is wrong with the sensor circuit. Please note that once SCR2 is activated, the complete circuit requires a reset either by switching off the power supply using switch S6 or pressing reset switch S1 for five seconds.

For better triggering of SCR1 and SCR2, resistors R37 and R29 can be replaced with 4.7-kilo-ohm each.

4. The internal connection diagram was not included because of limited space. The details are given below:

Terminal	Connection
1	Phase input 220V AC
2	Neutral input
3	Neutral output
4	Phase output RL1
	N/O contacts
5 and 6	DPDT relay N/O1
	contact for contactor
7 and 8	DPDT relay N/O2
	contact for capacitor
	point of starter button
9	Sensor A
10	Sensor B
11	Sensor C
12	Sensor D

5. The level sensors can be made with M4x 25mm long stainless-steel screws having two washers and nuts. The sensor wires can be twisted around the screw head and secured firmly with the help of washers and nuts. Apply some epoxy compound mix around the screw head to cover it completely, leaving 15mm length exposed so that the stainless-steel metal can come in contact with water to sense the water level. Ensure that the epoxy becomes hard before use.

For overhead tank flow sensor, a two-pin mains cord can be used. Fix it around the overhead tank's water-inflow pipe such that pins come in contact with water when it flows into the overhead tank.

One can also use a suitable-diameter plastic pipe and fix two overhead tank screws 180° apart diametrically (screw head inside the pipe and nuts from outside) to secure the sensing

wires. Secure the wires with the help of an epoxy compound. Fix this plastic pipe at the water inflow end of the pipe, so that water comes in contact with the sensors while flowing through the water pipe.

Solar Tracker

I have noticed an error in the 'Solar Tracker' circuit published in January issue. IC1 (LM339), which is a quad-comparator IC, doesn't give the required output as it is an open-collector op-amp. For the circuit to function properly, either four resistors have to be added between the output of each op-amp and supply rail, or the circuit has to be re-designed with LM324 comparator.

Arjun B.

Through e-mail

EFY: A 10-kilo-ohm pull-up resistor has to be used at each output of A1, A2, A3 and A4 in the circuit. The pull-up resistors were missed in the final print. Thanks for pointing out the mistake!

Electronic Calculator

In August 2011 issue, the 'Electronic Calculator' construction project is built around AT89C52 microcontroller. Can this microcontroller be replaced with P80C51BH microcontroller?

Santosh Kumar Bisoyee

Gunupur, Orissa

The author Dr D.G. Vyas replies:

I have designed the circuit using AT89C52. If P80C51BH and AT89C52 are similar, one can use it. But why not use AT89C52? I can send a readily designed simple kit for the calculator.

Robotic Arm

I am working on the 'Robotic Arm' construction project published in November 2011 issue. Please tell me the configuration bit setting of the microcontroller IC or send the source code.

Sujay Bonbagkar

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The author Venkat Raju replies: The reader may download the code from the link: <http://code.google.com/p/efyte-robo-arm/>